Name
Date
Course

Pre-Algebra Test III

Instructions

i) As always, write a title, name, date, and course.
ii) Number each problem, circle the number, and show all work.
iii) Draw a rectangle around your answers.
iv) Rewrite the equation whenever you perform an operation on both sides to preserve the result of the previous step.

Solve.

1. $8x+5=-2x$
2. $-3+4y-6y=-10y+1$
3. $-7=3(2z-1)+5$
4. $1-3x=-6x-x$
5. $-(x-3)+1=7x$
6. $-z+10z=-2(-4z+1)$
7. $-5-3x=-8+x$

Evaluate.

8. $2x+1,\ x=-5$
9. $-5y^2-6,\ y=-1$
10. $\frac{x}{-3}-5,\ x=27$
11. $\frac{1-x}{2},\ x=-7$
12. $-2(2x-5)^2-4,\ x=2$
13. $2y^2-5y+1,\ y=0$

⑭ $1-3(-4z-1)^2, z=-1$ ⑮ a) $-x^2, x=-10$ b) $-y^2, y=9$

c) $z^2, z=-6$ d) $x^3, x=-1$

⑯ a) $-7z^2, z=2$ b) $-3x^2, x=-5$ ⑰ a) $2y^3, y=-3$ b) $-z^3, z=1$

⑱ $2(5x+6)-6, x=4$ ⑲ $8-10(y-2)^2+\frac{3+2y}{5-y}, y=1$

⑳ a) Find D if $D=b^2-4ac$, $b=-3$, $a=1$, and $c=-1$.

b) Find V if $V=LWH$, $L=10$, $W=6$, and $H=3$.

㉑ a) Find A if $A=\pi r^2$ and $\pi=3.14$ and $r=1$.

b) Find P if $P = \frac{nRT}{V}$ and $n=3$, $R=4$, $T=1$, and $V=6$.

Check the solutions below. If correct, write "True." If incorrect, write "False." Show the work that leads to your answer.

22 a) $x = 5x - 40,\ x = 10$ b) $5y + y = 4 - 2y + 4y,\ y = 1$

23 a) $5z - 24 = -8z + z,\ z = 2$ b) $2x - 5x - 9 = -2x - 2x,\ x = 0$

Solve.

24 $-(z+5) + 2z - 2(3z+1) = 10z + 3(2 - 4z)$

25 $5y - 8y + 6 = -2(-3y-1) - 4(2y+2)$

Pre-Algebra Test III Answers

1. $\boxed{x = -1/2}$
2. $\boxed{y = 1/2}$
3. $\boxed{z = -3/2}$
4. $\boxed{x = -1/4}$
5. $\boxed{x = 1/2}$
6. $\boxed{z = -2}$
7. $\boxed{x = 3/4}$
8. $\boxed{-9}$
9. $\boxed{-11}$
10. $\boxed{-14}$
11. $\boxed{4}$
12. $\boxed{-6}$
13. $\boxed{1}$
14. $\boxed{-26}$
15. a) $\boxed{-100}$ b) $\boxed{-81}$ c) $\boxed{36}$ d) $\boxed{-1}$
16. a) $\boxed{-28}$ b) $\boxed{-75}$
17. a) $\boxed{-54}$ b) $\boxed{-1}$
18. $\boxed{46}$
19. $\boxed{-3/4}$
20. a) $\boxed{D=13}$ b) $\boxed{V=180}$
21. a) $\boxed{A=3.14}$ b) $\boxed{P=2}$
22. a) $\boxed{10=10}$ $\boxed{\text{True}}$

    b) $\boxed{6=6}$ $\boxed{\text{True}}$
23. a) $\boxed{-14=-14}$ $\boxed{\text{True}}$

    b) $\boxed{-9=0}$ $\boxed{\text{False}}$
24. $\boxed{z = -13/3}$
25. $\boxed{y = 12}$